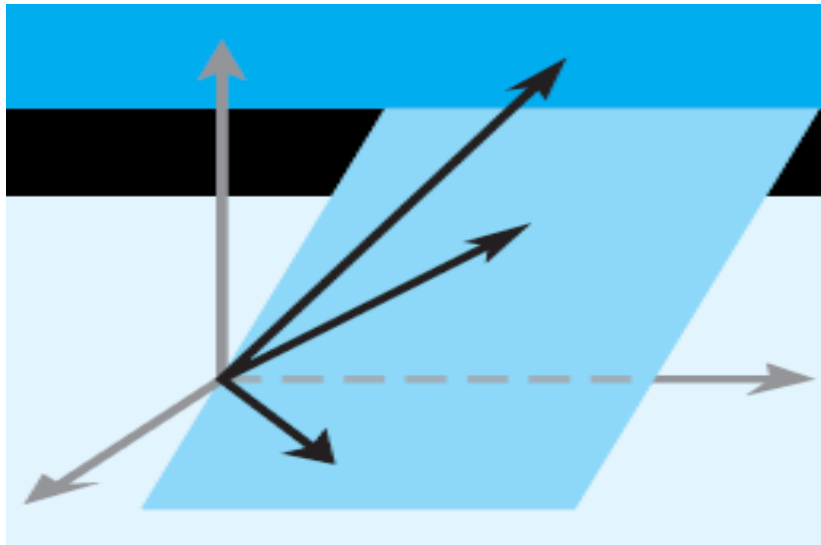


Elementary Linear Algebra

Chapter 7



Howard Anton
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Chapter 7

Diagonalization & Quadratic Forms

- **7.1 Orthogonal Matrices** Page 389
- **7.2 Orthogonal Diagonalization**
- **7.3 Quadratic Forms**
- **7.4 Optimization Using Quadratic Forms**
- **7.5 Hermitian, Unitary, and Normal Matrices**

Section 7.1 Orthogonal Matrices

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DEFINITION 1 A square matrix A is said to be *orthogonal* if its transpose is the same as its inverse, that is, if

$$A^{-1} = A^T$$

or, equivalently, if

$$AA^T = A^T A = I \quad (1)$$

Section 7.1 Orthogonal Matrices

► EXAMPLE 1 A 3×3 Orthogonal Matrix

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The matrix

$$A = \begin{bmatrix} \frac{3}{7} & \frac{2}{7} & \frac{6}{7} \\ -\frac{6}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{6}{7} & -\frac{3}{7} \end{bmatrix}$$

is orthogonal since

Columns of A are orthonormal vectors

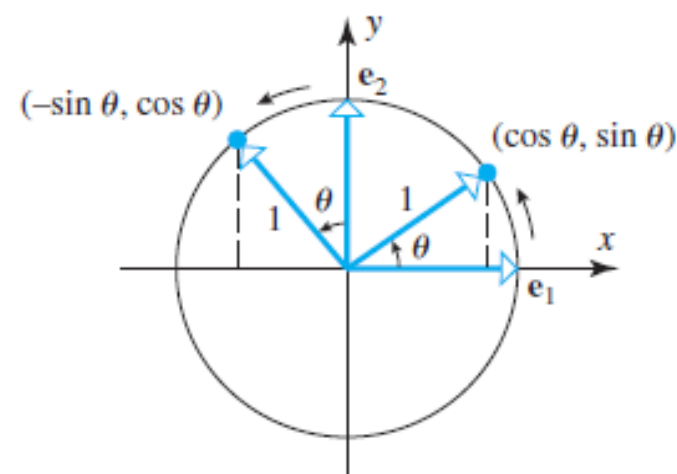
$$A^T A = \begin{bmatrix} \frac{3}{7} & -\frac{6}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{3}{7} & \frac{6}{7} \\ \frac{6}{7} & \frac{2}{7} & -\frac{3}{7} \end{bmatrix} \begin{bmatrix} \frac{3}{7} & \frac{2}{7} & \frac{6}{7} \\ -\frac{6}{7} & \frac{3}{7} & \frac{2}{7} \\ \frac{2}{7} & \frac{6}{7} & -\frac{3}{7} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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► EXAMPLE 2 Rotation and Reflection Matrices Are Orthogonal

Recall from Table [5](#) of Section [4.9](#) that the standard matrix for the counterclockwise rotation of $\mathbb{R}^2$ through an angle θ is

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



This matrix is orthogonal for all choices of θ since

$$A^T A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

We leave it for you to verify that the reflection matrices in Tables [1](#) and [2](#) and the rotation matrices in Table [6](#) of Section [4.9](#) are all orthogonal.

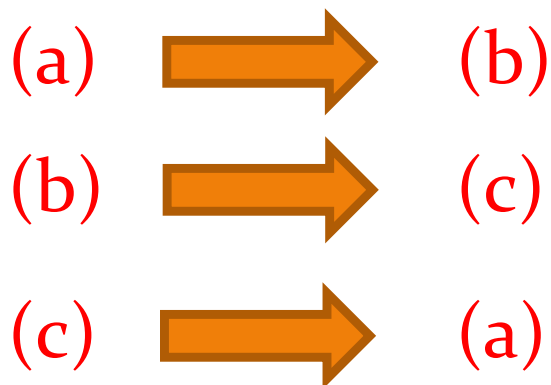
Section 7.1 Orthogonal Matrices

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THEOREM 7.1.1 *The following are equivalent for an $n \times n$ matrix A .*

- (a) *A is orthogonal.*
- (b) *The row vectors of A form an orthonormal set in R^n with the Euclidean inner product.*
- (c) *The column vectors of A form an orthonormal set in R^n with the Euclidean inner product.*

Proof



Section 7.1 Orthogonal Matrices

THEOREM 7.1.2

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- (a) *The transpose of an orthogonal matrix is orthogonal.*
- (b) *The inverse of an orthogonal matrix is orthogonal.*
- (c) *A product of orthogonal matrices is orthogonal.*
- (d) *If A is orthogonal, then $\det(A) = 1$ or $\det(A) = -1$.*

} *Exercises*

Proof

by Definition

(a) A is orthogonal matrix $\Rightarrow A^T = A^{-1}$

$\Rightarrow (A^T)^T = (A^{-1})^T$

$\Rightarrow = (A^T)^{-1}$ **Theorem 1.4.9**

Section 7.1 Orthogonal Matrices

► **EXAMPLE 3** $\det(A) = \pm 1$ for an Orthogonal Matrix A

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The matrix

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

is orthogonal since its row (and column) vectors form orthonormal sets in $\mathbb{R}^2$ with the Euclidean inner product. We leave it for you to verify that $\det(A) = 1$ and that interchanging the rows produces an orthogonal matrix whose determinant is -1 .

Section 7.1 Orthogonal Matrices

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THEOREM 7.1.3 *If A is an $n \times n$ matrix, then the following are equivalent.*

- (a) A is orthogonal.
- (b) $\|A\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$ in R^n .
- (c) $A\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}$ and $\mathbf{y}$ in R^n .

Proof (a)  (b) $\|A\mathbf{x}\|^2 = \langle A\mathbf{x}, A\mathbf{x} \rangle \quad \because \quad A\mathbf{x} \in R^n$

$$\begin{aligned} &= (A\mathbf{x})^T A\mathbf{x} \quad \text{by dot product} \\ &= \mathbf{x}^T A^T A\mathbf{x} \\ &= \mathbf{x}^T \mathbf{x} \\ &= \|\mathbf{x}\|^2 \end{aligned}$$

Section 7.1 Orthogonal Matrices

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THEOREM 7.1.3 *If A is an $n \times n$ matrix, then the following are equivalent.*

- (a) A is orthogonal.
- (b) $\|A\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}$ in R^n .
- (c) $A\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}$ and $\mathbf{y}$ in R^n .

Proof

(b)



(c)

Exercises

(c)



(a)

Exercises

Section 7.1 Orthogonal Matrices

Section 4.9 Matrix Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$

THEOREM 4.9.1 *For every matrix A the matrix transformation $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ has the following properties for all vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and for every scalar k :*

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(a) $T_A(\mathbf{0}) = \mathbf{0}$

(b) $T_A(k\mathbf{u}) = kT_A(\mathbf{u})$ [Homogeneity property]

(c) $T_A(\mathbf{u} + \mathbf{v}) = T_A(\mathbf{u}) + T_A(\mathbf{v})$ [Additivity property]

(d) $T_A(\mathbf{u} - \mathbf{v}) = T_A(\mathbf{u}) - T_A(\mathbf{v})$

THEOREM 4.9.2 *If $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T_B: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are matrix transformations, and if $T_A(\mathbf{x}) = T_B(\mathbf{x})$ for every vector $\mathbf{x}$ in $\mathbb{R}^n$, then $A = B$*

Section 7.1 Orthogonal Matrices

Recall: Section 4.4 Coordinates and Basis

DEFINITION 2 If $S = \{v_1, v_2, \dots, v_n\}$ is a basis for a vector space V , and

$$v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

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is the expression for a vector v in terms of the basis S , then the scalars $c_1, c_2, \dots, c_n$ are called the *coordinates* of v relative to the basis S . The vector $(c_1, c_2, \dots, c_n)$ in R^n constructed from these coordinates is called the *coordinate vector of v relative to S* ; it is denoted by

$$(v)_S = (c_1, c_2, \dots, c_n) \quad (6)$$

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THEOREM 7.1.4 *If S is an orthonormal basis for an n -dimensional inner product space V , and if*

$$(\mathbf{u})_S = (u_1, u_2, \dots, u_n) \quad \text{and} \quad (\mathbf{v})_S = (v_1, v_2, \dots, v_n)$$

$$\mathbf{u} \in V$$

$$\mathbf{v} \in V$$

$$u \in C[0,1]$$

then:

$$(a) \quad \|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2 + \dots + u_n^2}$$

$$(b) \quad d(\mathbf{u}, \mathbf{v}) = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$$

$$(c) \quad \langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

Section 7.1 Orthogonal Matrices

THEOREM 7.1.5 *Let V be a finite-dimensional inner product space. If P is the transition matrix from one orthonormal basis for V to another orthonormal basis for V , then P is an orthogonal matrix.*

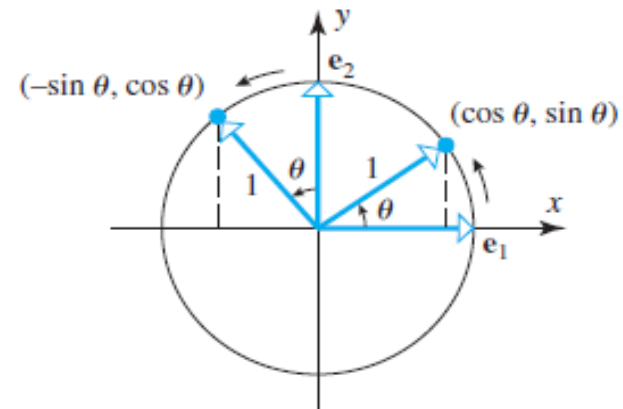
Proof **Page 395**

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► EXAMPLE 4 Rotation of Axes in 2-Space

In many problems a rectangular xy -coordinate system is given, and a new $x'y'$ -coordinate system is obtained by rotating the xy -system counterclockwise about the origin through an angle θ .

When this is done, each point Q in the plane has two sets of coordinates—coordinates (x, y) relative to the xy -system and coordinates (x', y') relative to the $x'y'$ -system (Figure 7.1.1a).



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► EXAMPLE 4 Rotation of Axes in 2-Space

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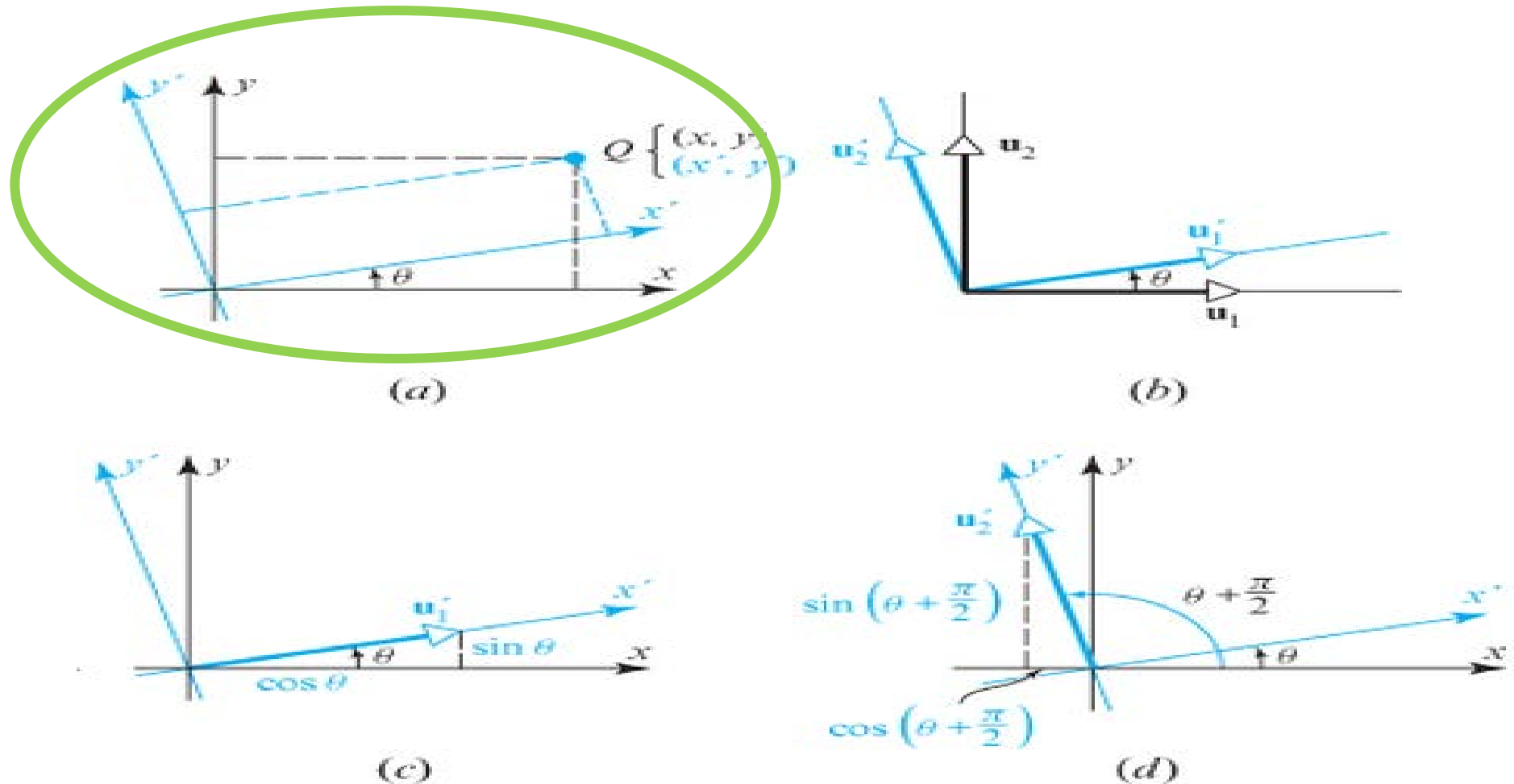


Figure 7.1.1

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► EXAMPLE 4 Rotation of Axes in 2-Space

By introducing unit vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ along the positive x - and y -axes and unit vectors $\mathbf{u}_1'$ and $\mathbf{u}_2'$ along the positive x' - and y' -axes, we can regard this rotation as a change from an old basis $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ to a new basis $B' = \{\mathbf{u}_1', \mathbf{u}_2'\}$ (Figure 7.1.1b). Thus, the **new** coordinates (x', y') and the **old** coordinates (x, y) of a point Q will be related by

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = P^{-1} \begin{bmatrix} x \\ y \end{bmatrix}$$

where P is the transition from B' to B .

Section 7.1 Orthogonal Matrices

► EXAMPLE 4 Rotation of Axes in 2-Space

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To find $\mathbf{P}$ we must determine the coordinate matrices of the new basis vectors $\mathbf{u}_1'$ and $\mathbf{u}_2'$ relative to the old basis.

As indicated in Figure 7.1.1 c, the components of $\mathbf{u}_1'$ in the old basis are $\cos \theta$ and $\sin \theta$, so

$$[\mathbf{u}_1']_B = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

Similarly, from Figure 7.1.1d we see that the components of $\mathbf{u}_2'$ in the old basis are $\cos(\theta + \pi/2) = -\sin \theta$ and $\sin(\theta + \pi/2)$, so

$$[\mathbf{u}_2']_B = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$$

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Thus the transition matrix from B' to B is

$$P = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (3)$$

Observe that P is an orthogonal matrix, as expected, since B and B' are orthonormal bases. Thus

$$P^{-1} = P^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

so (2) yields

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

P has the effect of rotating the xy -axes of a rectangular coordinate system through an angle θ .
(4)

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so (2) yields

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (4)$$

or, equivalently,

$$\begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \end{aligned}$$

These are sometimes called the *rotation equations* for $\mathbf{R}^2$.

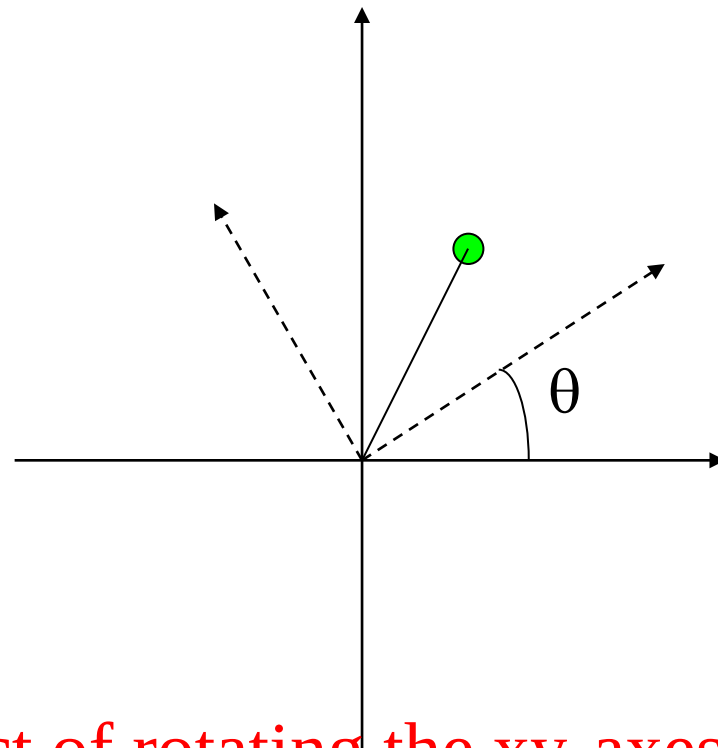
Section 7.1 Orthogonal Matrices

Rotated Coordinate System

$$\tilde{x}_1 = x_1 \cos \theta + x_2 \sin \theta$$

$$\tilde{x}_2 = -x_1 \sin \theta + x_2 \cos \theta$$

$$\begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



P has the effect of rotating the xy -axes of a rectangular coordinate system through an angle θ .

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► EXAMPLE 5 Rotation of Axes in 2-Space

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Use form [4](#) of the rotation equations for R^2 to find the new coordinates of the point $Q(2, 1)$ if the coordinate axes of a rectangular coordinate system are rotated through an angle of $\theta = \pi/4$.

Solution

Since

$$\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

the equation in [4](#) becomes

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Thus, if the old coordinates of a point Q are $(x, y) = (2, -1)$, then

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$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{3}{\sqrt{2}} \end{bmatrix}$$

so the new coordinates of Q are $(x', y') = \left(\frac{1}{\sqrt{2}}, -\frac{3}{\sqrt{2}}\right)$.



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► EXAMPLE 6 Application to Rotation of Axes in 3-Space

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Suppose that a rectangular xyz -coordinate system is rotated around its z -axis counterclockwise (looking down the positive z -axis) through an angle θ (Figure 7.1.2). If we introduce unit vectors $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ along the positive x -, y -, and z -axes and unit vectors $\mathbf{u}'_1$, $\mathbf{u}'_2$, and $\mathbf{u}'_3$ along the positive x' -, y' -, and z' -axes, we can regard the rotation as a change from the old basis $B = \{ \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \}$ to the new basis $B' = \{ \mathbf{u}'_1, \mathbf{u}'_2, \mathbf{u}'_3 \}$. In light of Example 4, it should be evident that

$$[\mathbf{u}'_1]_B = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} \quad \text{and} \quad [\mathbf{u}'_2]_B = \begin{bmatrix} -\sin \theta \\ \cos \theta \\ 0 \end{bmatrix}$$

Moreover, since $\mathbf{u}'_3$ extends 1 unit up the positive z' -axis,

$$[\mathbf{u}'_3]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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It follows that the transition matrix from B' to B is

$$P = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and the transition matrix from B to B' is

$$P^{-1} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(verify). Thus, the new coordinates (x', y', z') of a point Q can be computed from its old coordinates (x, y, z) by

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

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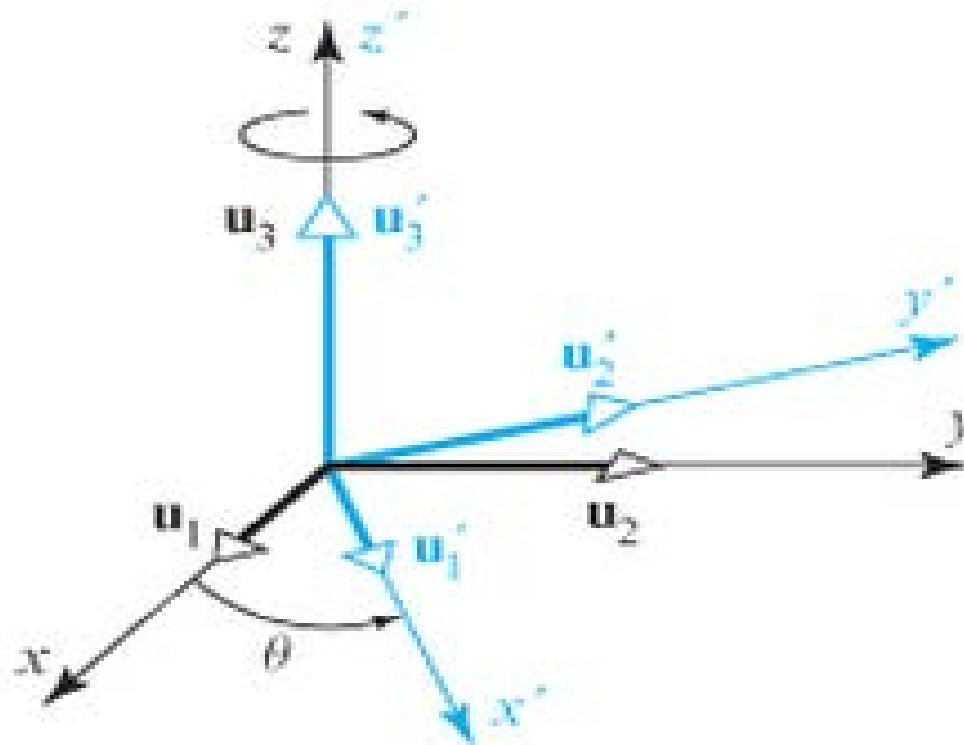


Figure 7.1.2

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Homework

Sec 7.1

prove Theorem 7.1.2

prove Theorem 7.1.3

True/False Page 395

1, 3 Page 426

4 Page 426

14, 15 (study) Page 426